CHAPTER 9

# Orthogonality and distances

S48, S51, S54, S56, S60

> **Key point · Objectives.** Calculate a dot product; justify orthogonality; find an orthogonal projection and a distance; calculate a volume.

## Course notes

The coordinate calculations in this chapter are carried out in an orthonormal frame of space.

### Dot product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their angle. If one of them is zero, the dot product is 0.

$$\vec{u} \cdot \vec{v} = \tfrac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal frame, for $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The dot product is symmetric and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their dot product is zero.

**Example.** $\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 - 2 + 0 = 0$. Their norms are 3 and $\sqrt{5}$.

### Line, plane and normal vector

Two lines are orthogonal if their direction vectors are. A line is orthogonal to a plane if it is orthogonal to two non-collinear directions of that plane. A normal vector to a plane is a non-zero vector orthogonal to all its directions.

**Why two directions suffice.** If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for every $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

> **Key point · Vector condition for belonging to a plane.** If $A$ belongs to the plane $\mathcal{P}$ with normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

### Orthogonal projection and distance

The orthogonal projection $H$ of $M$ onto a plane is the unique point of the plane such that $\overrightarrow{MH}$ is normal to the plane; if $M$ belongs to the plane, $H = M$. The distance from $M$ to the plane is $MH$. On a line with direction vector $\vec{u}$, the projection satisfies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

**Proof of the minimum.** For every point $P$ of the plane, $\overrightarrow{MH} \perp \overrightarrow{HP}$, so $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality holds only for $P = H$. The projection is the closest point of the plane.

**Method on a line.** If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and solve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Hence $t = \frac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

## Angles, areas and volumes

For an angle $\widehat{ABC}$, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos \widehat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{BA \times BC}$. For the acute angle between two lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}} h}{3}$, where $h$ is an orthogonal distance. The units are squared and cubed respectively.

## Expansions and loci

By bilinearity,

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

We can therefore also calculate $\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$. For two distinct points $A, B$ with midpoint $I$, the locus of points $M$ such that $MA = MB$ is the plane through $I$ with normal vector $\overrightarrow{AB}$: the *perpendicular bisector plane*. Indeed, writing the vectors from $I$, the expansions give $MA^2 - MB^2 = 2\overrightarrow{IM} \cdot \overrightarrow{AB}$. Equality of the distances is therefore equivalent to $\overrightarrow{IM} \cdot \overrightarrow{AB} = 0$.
**Extension, not assessed in 2026–2027.** Two planes are perpendicular when their normal vectors are orthogonal. This property must not be confused with parallelism of planes, which corresponds to collinear normal vectors.

## Research activity

**The point of a line closest to a given point — 35 min.**

In an orthonormal frame of space, consider $A(1,0,0)$, $M(0,2,3)$ and the line $d$ through $A$ with direction vector $\vec{u} = (1,1,0)$. We look for the point of $d$ closest to $M$.

1. Express, in terms of a real parameter $t$, the coordinates of a point $H$ of $d$.
2. Find the value of $t$ for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the resulting point $H$.
3. Calculate the distance $MH$.
4. Let $P$ be any point of $d$. Prove that $MP^2 = MH^2 + HP^2$, then deduce that $H$ is the unique point of $d$ closest to $M$.

## Progressive exercises

**E1 · Practice.** In an orthonormal basis of space, consider $\vec{u} = (2, -1, 3)$ and $\vec{v} = (1, 4, 0)$. Calculate $\vec{u} \cdot \vec{v}$.

**E2 · Practice.** In an orthonormal basis, consider $\vec{u} = (a, 2, 1)$ and $\vec{v} = (3, -1, 4)$, where $a$ is real. Find $a$ so that $\vec{u}$ and $\vec{v}$ are orthogonal.

**E3 · Practice.** In an orthonormal frame, a plane is spanned by $\vec{u} = (1, -1, 0)$ and $\vec{v} = (1, 0, -1)$. Show that $\vec{n} = (1, 1, 1)$ is normal to it. Then consider $A(0, 0, 0)$ and $B(2, 2, 2)$. Describe geometrically the set of points equidistant from $A$ and $B$.

**E4 · Practice.** In an orthonormal basis, consider $\vec{u} = (1, 0, 1)$ and $\vec{v} = (0, 1, 1)$. Find the measure, between 0 and $\pi$, of the angle between these two vectors.

**E5 · Practice.** A tetrahedron has as its base a right-angled triangle whose two perpendicular sides measure 3 cm and 4 cm. The height of the tetrahedron relative to this base is 5 cm. Calculate its volume in $\text{cm}^3$.

## Textbook or past-paper work

**Barbazo, p. 362, exercises 23 and 27.** In S48: exercise 23, to calculate a dot product in coordinates. At home after S54: exercise 27; check orthogonality to the two direction vectors of a plane, without settling for a single zero product.

## Homework

**HW11 — A face and its height — 50 min.**

In an orthonormal frame with origin $O$, consider the points $A(2,0,0)$, $B(0,2,0)$ and $C(0,0,2)$.

1. Prove that the triangle $ABC$ is equilateral and calculate its area.
2. Show that the vector $\vec{n} = (1,1,1)$ is normal to the plane $(ABC)$.
3. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Justify that its coordinates are of the form $(t,t,t)$. Find $t$ using $\overrightarrow{AH} \cdot \vec{n} = 0$.
4. Calculate $OH$, then the volume of the tetrahedron $OABC$ taking $ABC$ as the base. Check this volume by taking $OAB$ as the base.